

Ventilators: The primary purpose of ventilation is the control of the interior environment of the building by the removal / reduction of following:

- **Heat buildup**, thus providing comfort for workers, preserving goods and enabling equipment to function properly.
- **Gaseous by-products**, (the result of some manufacturing processes) thus providing a healthier atmosphere for workers.
- **Flammable fumes**, thus minimizing fire hazards (ventilation is also important after a fire has started at which time it helps in removing the fumes and smoke), and improving visibility for escapees and fire fighters.

Ventilation, whether natural or forced, does not have heat reduction characteristics. Air conditioning must be considered if a cooler temperature is desired.

Ventilation equipment comes in two categories:

- Inlet equipment
- outlet equipment

Either one can be natural or forced (power).

The planning and correct distribution of ventilation equipment has a major role in the overall efficiency of the ventilation system. Ventilation efficiency is also affected by the location of equipment, partitions and doors inside a building.

A detailed study of ventilation must be made at the planning stage of the building. For complex buildings, determining ventilation requirements may require vast calculations. Much simpler procedures normally suffice for typical metal buildings.

There are two simple and practical methods for determining ventilation requirements:

- **The Air Change Method**, which is based on a recommended rate of air changes per hour for different building usages when ventilator capacity is given.
- **The Heat Removal Method**, which calculates the volume of air required to remove the heat gain inside a building.

These methods are good tools to approximate the ventilation requirements of the building. More detailed and accurate ventilation plans should be sought by contracting ventilation specialists directly.

Table 1			
Stack Height (m)	Temperature Difference (°C)	R = Exhaust Capacity (m ³ /s)	
		MRV 300	MRV 600
3	5	0.779	1.559
6	5	0.916	1.832
	10	1.109	2.218
9	5	1.021	2.041
	10	1.257	2.514
	15	1.438	2.876
12	10	1.382	2.763
	15	1.591	3.182
15	10	1.492	2.983
	15	1.726	3.452

Ventilation Design Using Air Change Method

$$Q_v = \frac{V \times N}{R \times 3600} \text{ where,}$$

Q_v = Quantity of ventilators.

R = Exhaust capacity (m^3/sec).

V = Building volume (m^3).

N = Air change per hour.

In Table 1, enter the stack height, which is the average of the eave height and the ridge (peak) height. Enter the temperature difference between the inside and outside of the building. Read the equivalent required total exhaust capacity in m^3/sec .

The number of recommended air changes per hour is obtained from Table 2.

Finally, apply the aforementioned equation to determine the number of required ventilators. For efficient functioning of the ventilation system the free inlet area (permanent openings plus the effective area of the louvers) must be greater than 150% of the ventilation area.

Table 2		
Type of Building	N = Recommended Air Changes Per Hour	
	From	To
Warehouses, factories, dining rooms, machine shops, engine rooms, textile mills, wood working shops	5	10
Boiler rooms, paint shops, garages, schools	10	15
Buildings with fumes, kitchens, paper mills, textile mills, dye houses	15	20



Ventilation Design Using Heat Removal Method

Calculate the total heat gain (H) in the building:

$$H = H_s + H_l + H_i \quad \text{where,}$$

H_l = Heat gain from lighting

H_i = Heat gain from internal equipment, people, etc.

H_s = Solar Heat Gain

$$H_s = A \times I \times a \times e \quad \text{where,}$$

A = Roof Area

I = Actual Solar Radiation striking the roof surface (=1.12 kW/m² for UAE)

a = Solar radiation factor of the sheets (=0.40 for Mammut standard roof sheeting)

e = Proportion of "a" transmitted into the building. This factor is dependent on the "U" value of insulation

= 0.093 for a building with insulation

= 1.000 for a building without insulation

Calculate the ventilation rate required ($VR_{req'd}$):

$$VR_{req'd} = \frac{H}{S_H \times D \times T} \quad \text{where,}$$

S_H = Specific heat capacity of air
= 1.005

D = Density of air = 1.206 kg/m³

T = Temperature difference (use °C difference per meter of stack height).

Determine the quantity of ventilators needed,

Quantity of ventilators

$$= \frac{VR_{req'd}}{R}$$

where,

R = Exhaust capacity in m³/s of all ventilators (see page 2 of 13 of this **section**)

Calculate the total area of gravity ventilators:

Total ventilator area

$$= \frac{VR_{req'd}}{[0.03 \times \text{stack height} \times T]^{1/2}}$$

For special ventilators, the supplier's catalogue must be consulted.



Example of Ventilation Design using the Air Change Method

A Clear Span building that is used as a factory requires ventilation using gravity ridge ventilators. The building is 140 m long and 36 m wide. The eave height is 9 m. The roof has a 0.5/10 slope.

The temperature difference between the inside and outside of the building is not expected to exceed 10°C. Determine the number and size of gravity ventilators that are needed.

The average of the eave height and the ridge height.	Stack Height	= 9.45 m
Exhaust capacity (R) at 9.45 m stack height is determined by interpolation from the table on page 1 of 13 of this section.	R (for MRV 300) R (for MRV 600)	= 1.276 m ³ /s = 2.551 m ³ /s
The recommended number of air changes per hour (N) for factories is 5 to 10 times.	N	= 8
Volume of building.	V	= 140 x 36 x 9.45 = 47628 m ³
Quantity of MRV-300 ventilators required. Need 83 MRV-300 ventilators	Q _v	= $\frac{(47628 \times 8)}{1.276 \times 3600}$ = 82.32 ~ 83
The maximum number of ridge ventilators (Q_{max}) that the building can accommodate. Q_{max} = $\frac{\text{Building Length (m)} - 1}{\text{Ventilator Length (m)}}$	Q _{max}	= $\frac{140-1}{3}$ = 46.33 ~ 46
MRV-300 Cannot be used. Try MRV-600	Q _{max} < Q _v	= 46 < 83
Quantity of MRV-600 ventilators required. Q_v < Q_{max} = 46	Q _v	= $\frac{(47628 \times 8)}{2.551 \times 3600}$ = 41.49 ~ 42
Use 42 gravity ridge ventilators MRV-600 for the building		

Gravity Ventilators

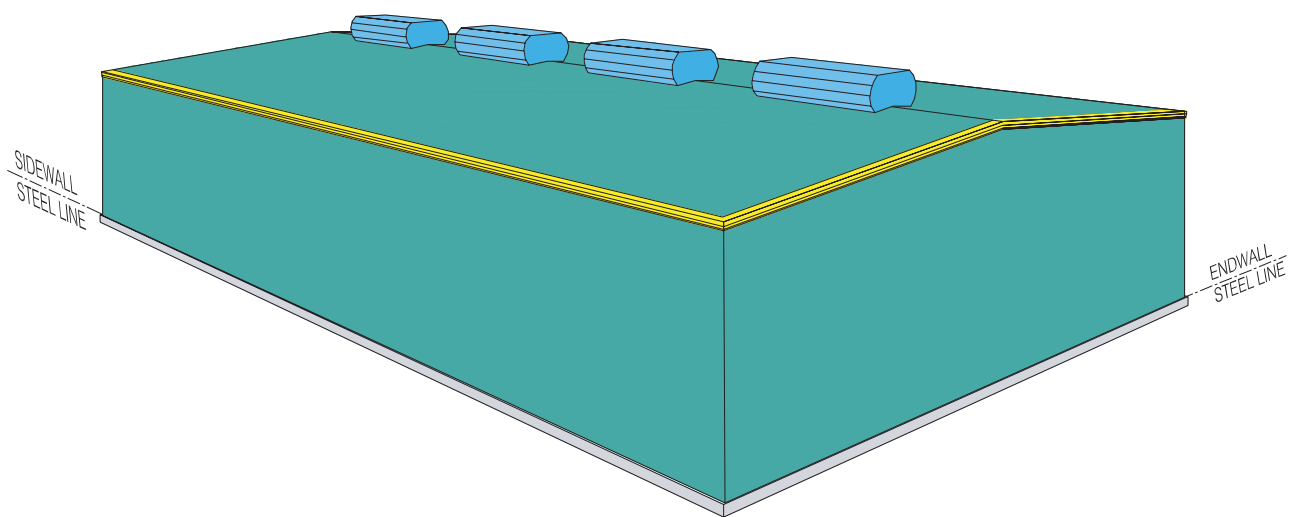
Mammut ridge ventilators will provide natural air circulation in any metal building.

Ridge ventilators for Mammut metal buildings shall be MRV 300 or MRV 600 and can be installed as continuous or single units. They are 3000 mm long units with fixed throat openings for gravity air flow.

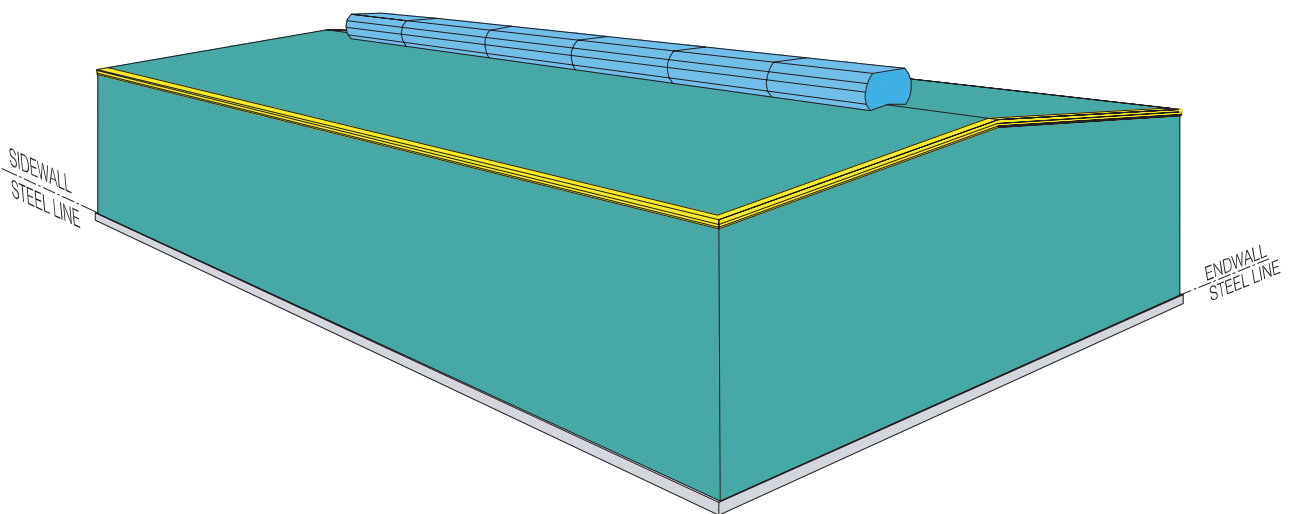
Ridge ventilators are designed for mounting to the building ridge using gravity flow to exhaust heat, fumes, and smoke at rapid rates and regulate free air within the building. Adequate air inflow must be provided to ensure proper ventilation.



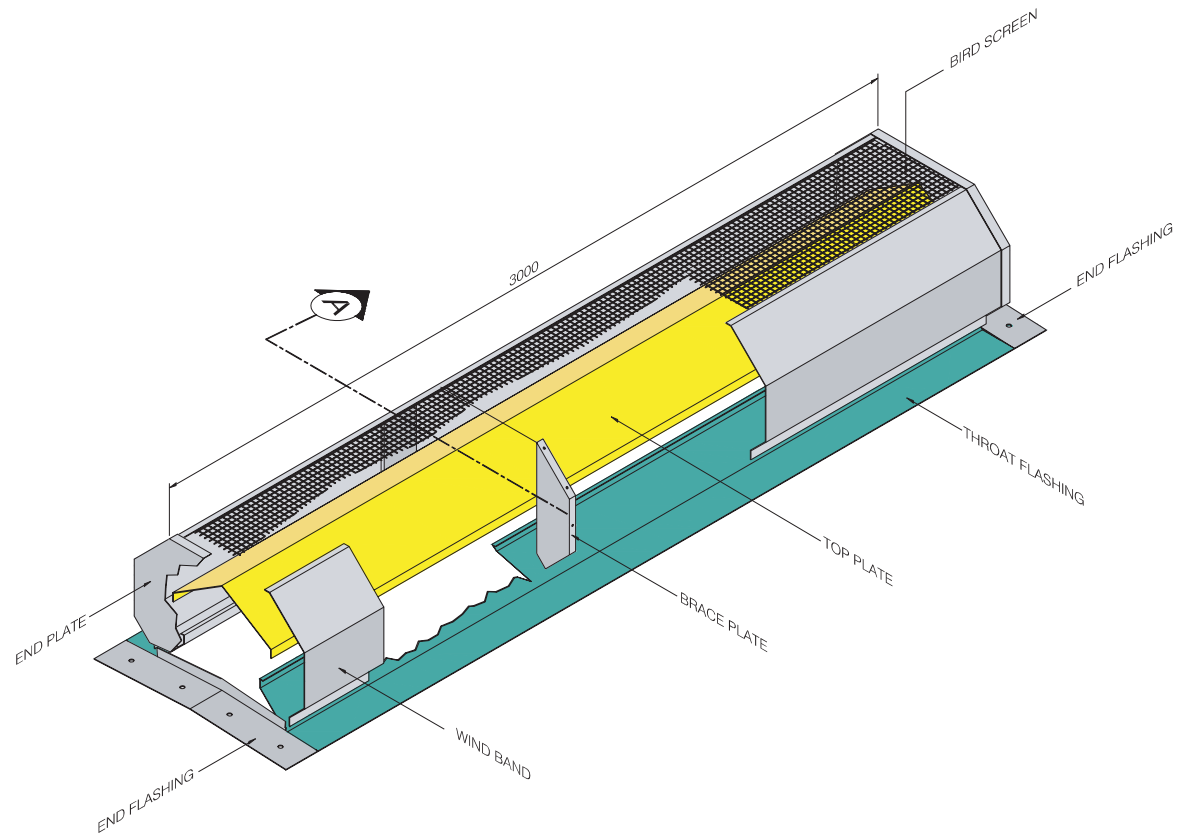
Specifications for Gravity Ridge Ventilators		
Type	MRV 300	MRV 600
Main Parts	Wind bands, top plate, throat flashing, end flashing, (all 0.5 mm pre-painted galvanized steel)	
Structure	Wind band (cold-formed plain sheet) with brace plates (0.5 mm pre-painted) and throat gussets (1.3 mm galvanized)	Wind band (cold-formed Profile 'C' panel) with framing*, that consist of cold-formed channels and hot rolled angles
Length	3000 mm	
Installation	At ridge, continuous units or single units	
Throat	300 mm	600 mm
Throat Area	0.9 m ²	1.8 m ²
Throat Opening	Fixed	
Damper	Option available	Option not available
Bird Screen	1.06 mm galvanized 12 mm x 12 mm mesh	2.3 mm galvanized 16 mm x 16 mm mesh
Finish	Any Mammut standard can be supplied upon request.	



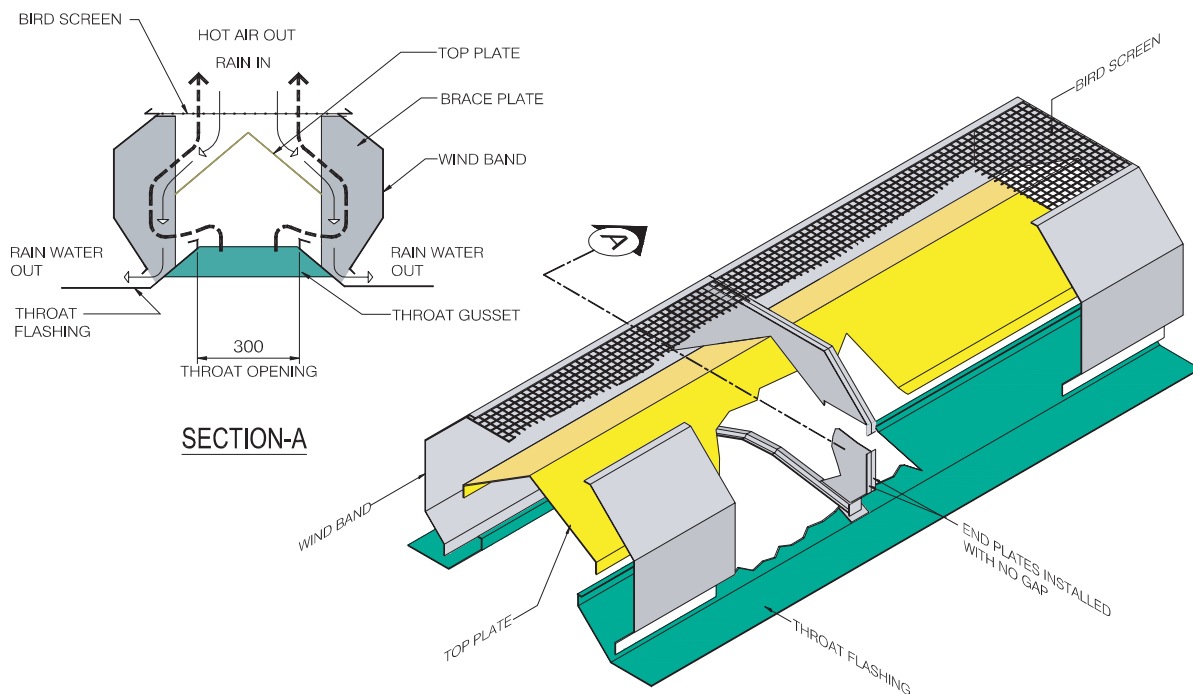
SINGLE RIDGE VENTILATOR



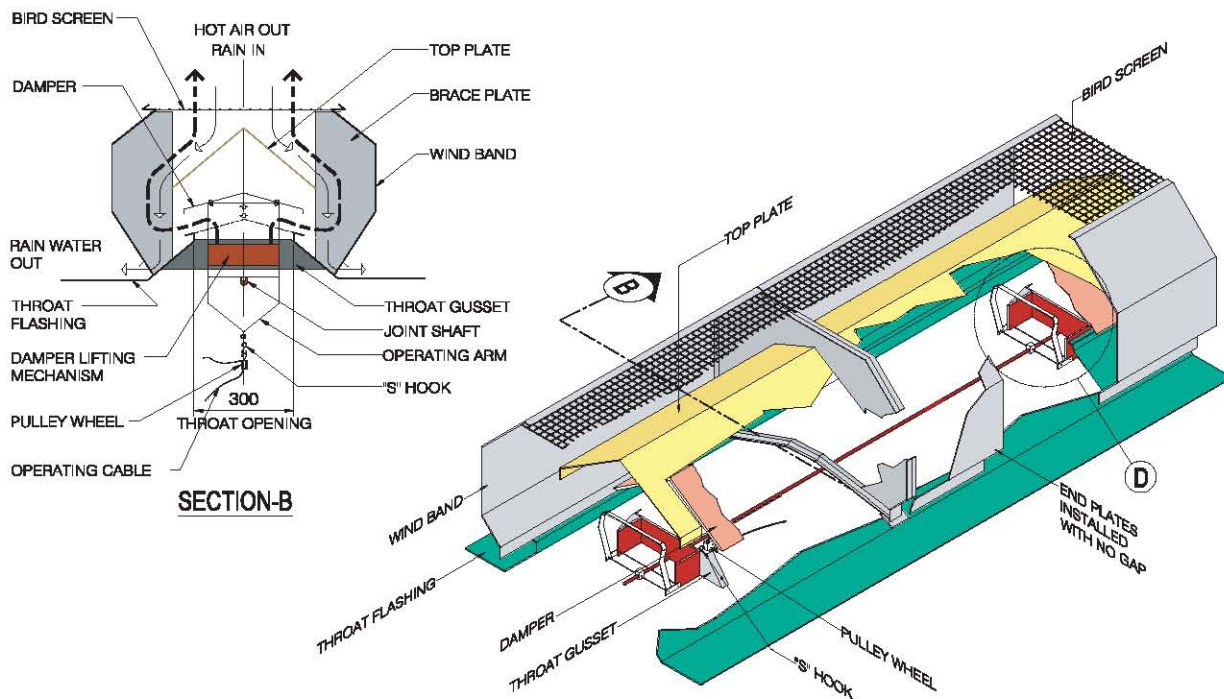
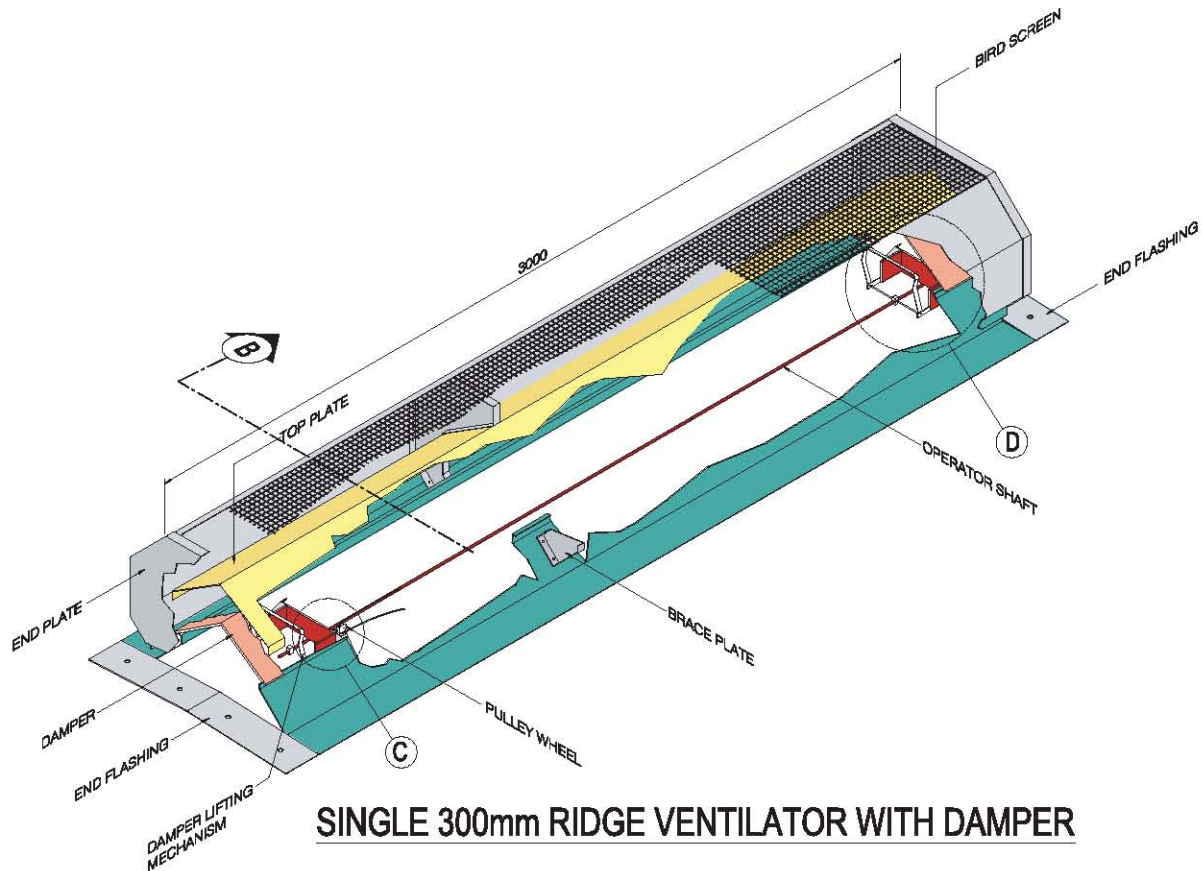
CONTINUOUS RIDGE VENTILATOR



SINGLE 300mm RIDGE VENTILATOR WITHOUT DAMPER



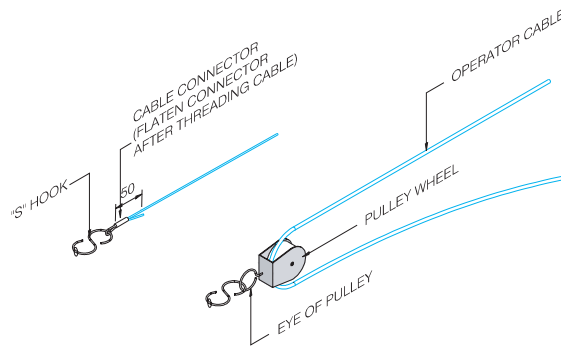
CONNECTION DETAILS AT CONTINUOUS RIDGE VENTILATOR



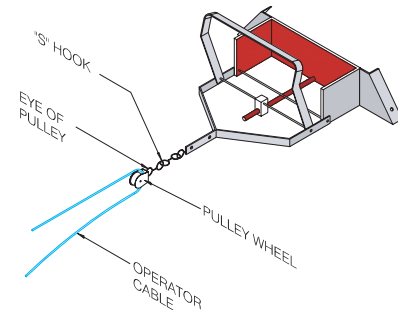
CHAPTER 13 : ACCESSORIES

SECTION 13.7 : Ventilators

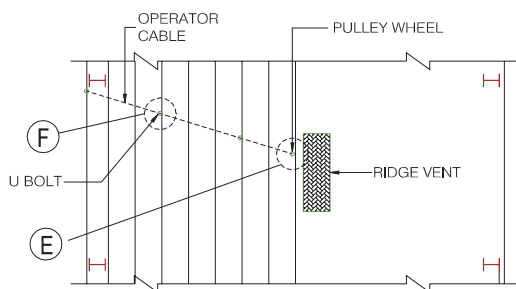
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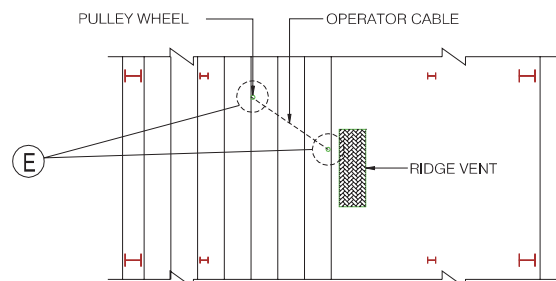
DETAIL-C



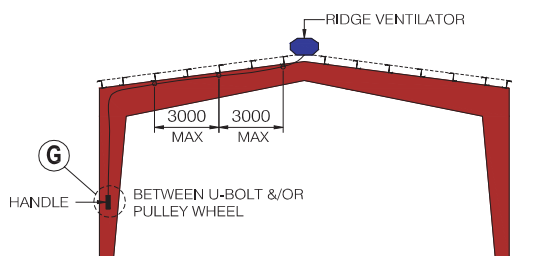
DETAIL-D



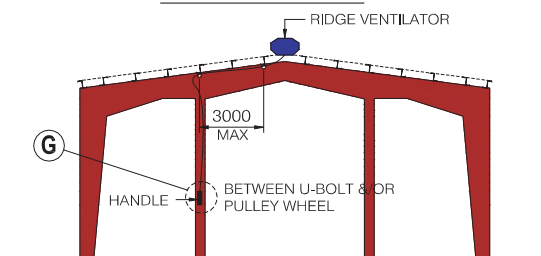
BUILDING PLAN



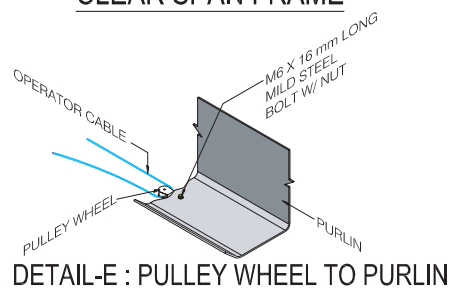
BUILDING PLAN



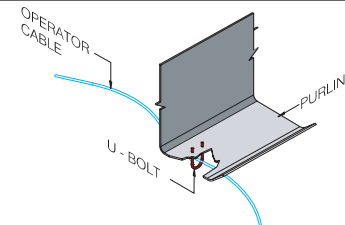
CLEAR SPAN FRAME



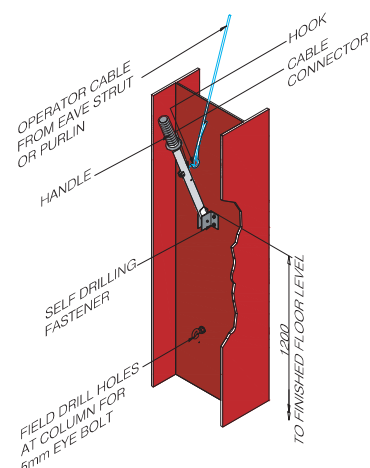
MULTI-SPAN FRAME



DETAIL-E : PULLEY WHEEL TO PURLIN

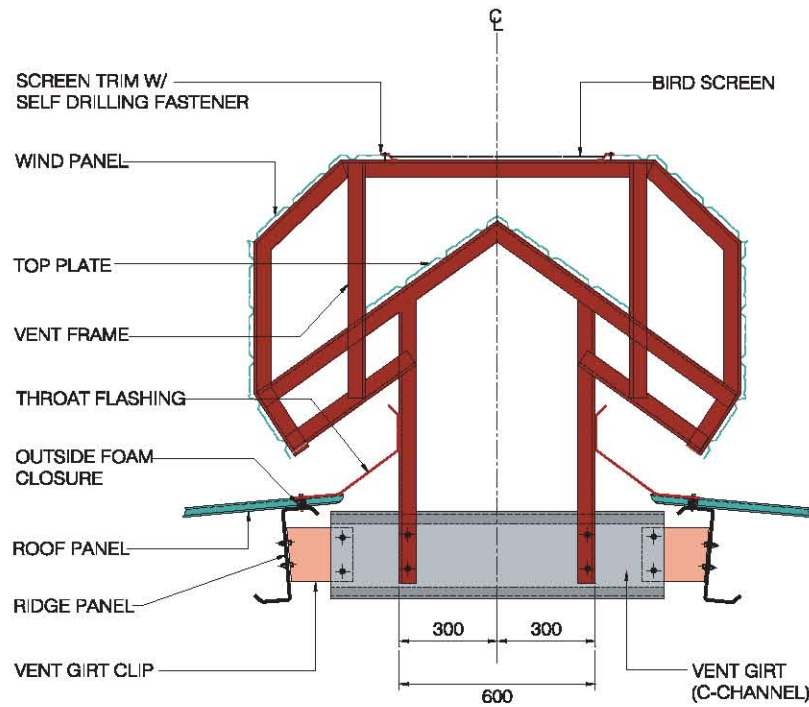


DETAIL-E : U BOLT TO PURLIN

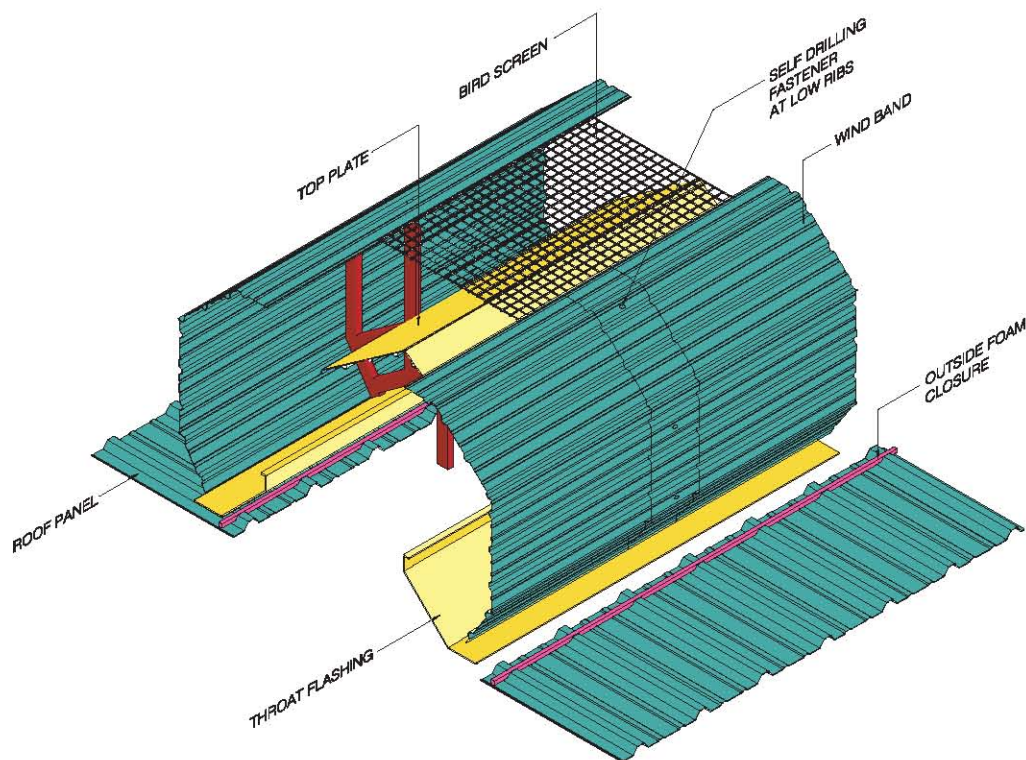


DETAIL-G : VENTILATOR HANDLE TO COLUMN

RIDGE VENTILATOR OEPRATION DETAILS



THROAT OPENING



600mm THROAT RIDGE VENTILATOR (SINGLE OR CONTINUOUS)

Power Ventilators

Power ventilators consist of a one-piece aluminum base and throat assembly complete with spun aluminum non-return shutters incorporating an anti-dust/sand seal.

Power is supplied by means of a direct drive motor with an integral aluminum aerofoil blade impeller, statically and dynamically balanced as a complete unit. The motor and impeller are fitted with guards for protection against moving parts. Noise level does not exceed 66 dB(A) at a distance of 2.0 m. The unit is weatherproof against rain and dust, considering a maximum rainfall of 75 mm/h and wind speed of 130 km/hour. The power unit is fitted with a non return shutter, which closes when the unit is not in operation and is held in position by a centrifugal locking mechanism.

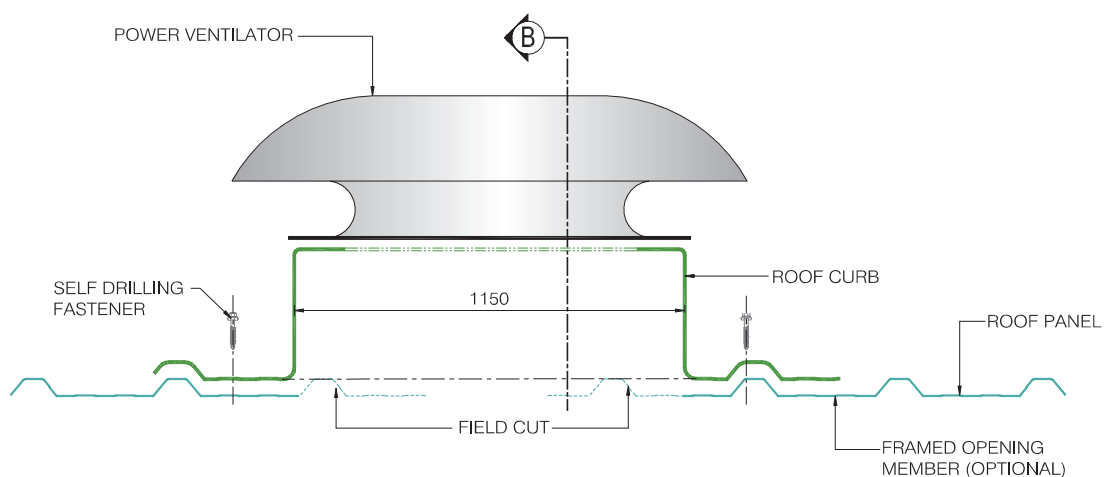
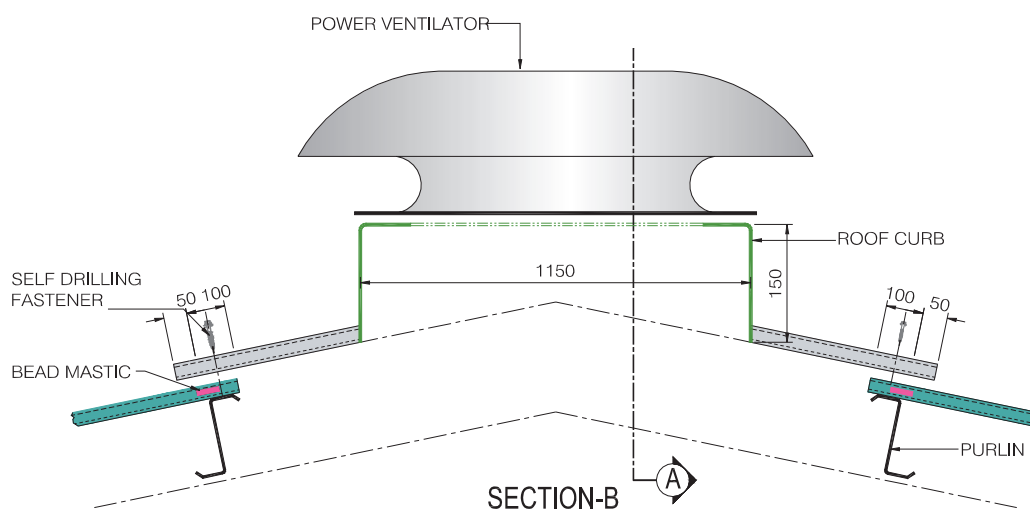
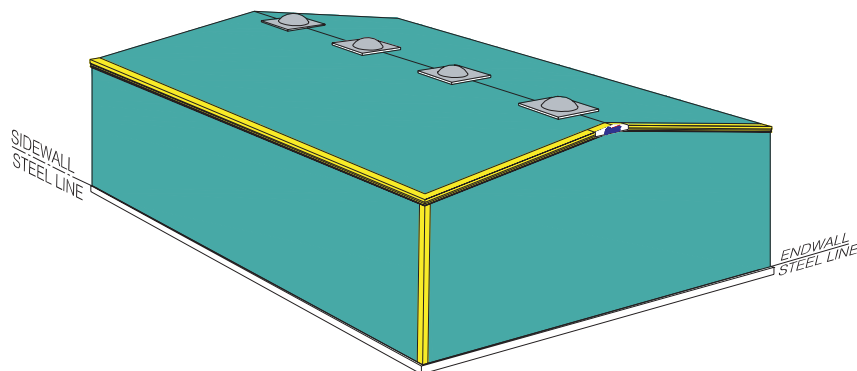
Recommended free air inlet area to be at least 2.5 m² per fan.

Ventilation design using the Heat Removal Method is as illustrated on page 3 of 13 of this section

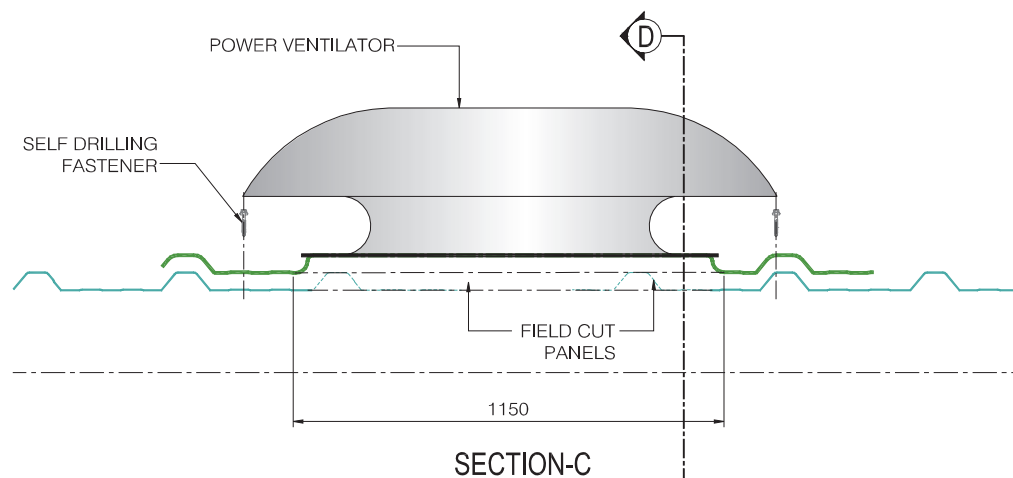
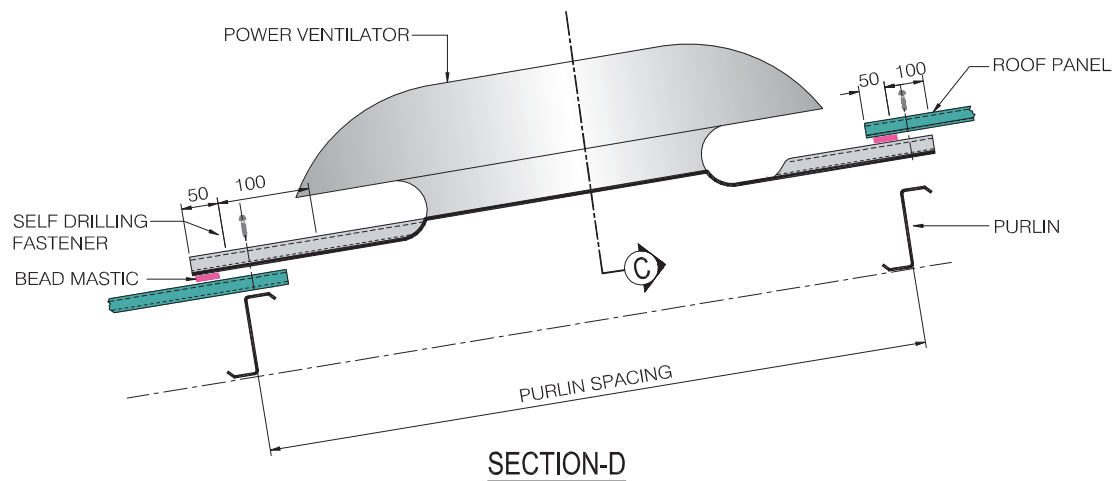
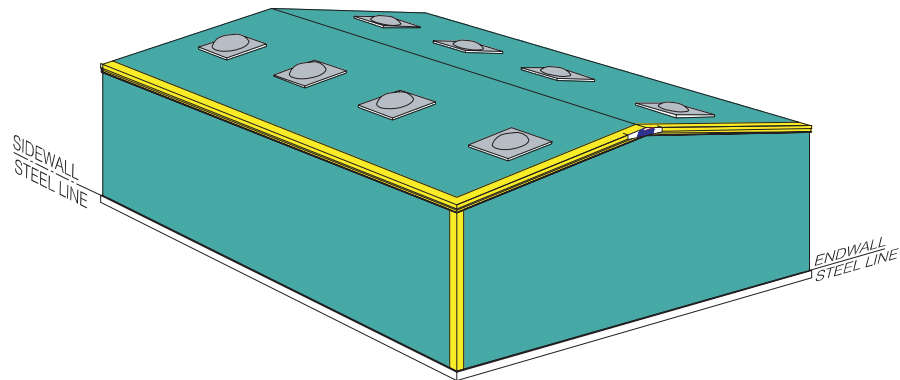
For special ventilators, the supplier's catalogue must be consulted.



Power Ventilator [Colt Tornado T630] (Weight = 32 kg)			
Performance Range Table			
Power Supply			Exhaust Volume (m ³ /sec)
Phase	Frequency (Hz)	Voltage (V)	
3	50	415	2.4
	60	380	2.6



POWER VENTILATOR AT ROOF RIDGE



POWER VENTILATOR AT ROOF SLOPE